

Industrial Electrical System Module 1

2 Marks Questions & Answers

Theory Questions

1. **Generation, transmission, distribution, use and trading of Electrical energy are governed by which Act in India?**

The electrical sector in India is governed by **The Electricity Act, 2003**. This act consolidates the laws relating to generation, transmission, distribution, trading, and use of electricity.

2. **Is there any license required for generation of Electricity in India?**

According to The Electricity Act, 2003, **no license is required** for the generation of electricity (excluding hydro-power projects involving defined capital investment and nuclear power), provided the generating station complies with the technical standards for grid connectivity.

3. **How many years a license is valid for the distribution of electricity?**

A distribution license is typically valid for a period of **25 years**, unless revoked earlier by the appropriate regulatory commission.

4. **What is the maximum value of High Voltage as per Indian Electricity Rule?**

As per the Indian Electricity Rules (1956), "High Voltage" is defined as voltage exceeding 650 Volts but not exceeding 33,000 Volts. Therefore, the maximum value is **33,000 Volts (33 kV)**.

5. **What is the declared voltage and allowable variation for a single-phase low voltage AC supply in our country?**

The standard declared voltage is **230V**. As per IS regulations and IE Rule 54, the allowable variation for low voltage is $\pm 6\%$ of the declared voltage.

6. **Name two energy intensive industries for which Energy Audit is compulsory?**

Two energy-intensive industries are:

- Iron and Steel Industry
- Cement Industry

7. **Name two types of Electrical diagram standardized by BIS.**

Two types of electrical diagrams standardized by the Bureau of Indian Standards (BIS) are:

- Circuit (Schematic) Diagram

- Wiring (Connection) Diagram

8. **What are the purposes of wiring diagram?**

A wiring diagram shows the physical layout of the electrical connections. Its purposes include aiding in the **installation** of wiring, showing the **routing** of cables between components, and facilitating **troubleshooting** by showing physical locations.

9. **What are the purposes of Circuit Schematic diagram?**

A schematic diagram represents the circuit logic using standard symbols. Its purposes are to explain the **principle of operation**, show the **electrical relationship** between components without regarding physical position, and assist in **design analysis**.

10. **Between Joint box system and Loop-in system of wiring, which one is less costly?**

The **Joint Box system** is generally considered less costly in terms of cable consumption because it requires less wire length compared to the Loop-in system, although it may be more labor-intensive.

11. **What is the function of braking magnet in an energy meter?**

The braking magnet provides a **controlling torque** (braking torque) by inducing eddy currents in the rotating disc. This ensures that the speed of the disc is directly proportional to the power consumed and prevents the disc from spinning when the load is removed.

12. **What is the function of Shading coil in an Energy meter?**

The shading coil creates a phase displacement (lag) in the magnetic flux of the pressure coil. Its function is to ensure that the flux from the pressure coil lags the supply voltage by exactly 90° , which is necessary for accurate power measurement at different power factors.

13. **What is the reservation in the use of Re-wirable Fuse?**

The main reservations are its **low breaking capacity** and **inaccurate calibration**. The fusing factor is unreliable (subject to oxidation and human error in wire selection), and it cannot safely interrupt high short-circuit currents compared to HRC fuses or MCBs.

14. **What is the maximum capacity of MCB available in the market?**

Miniature Circuit Breakers (MCBs) are typically available up to a maximum current rating of **100 Amperes (100A)** or sometimes 125A. Beyond this rating, Moulded Case Circuit Breakers (MCCBs) are used.

15. **What is the role of Arc chute in a MCB?**

The arc chute consists of a stack of metal plates (splitters). Its role is to **split, cool, and lengthen** the electric arc generated during contact separation, thereby extinguishing it quickly and safely.

16. **Name two applications of Two-way Switch.**

Two common applications are:

- **Staircase wiring:** Controlling a light from both the top and bottom of the stairs.
 - **Corridor lighting:** Controlling lights from either end of a long hallway.
17. **Why RCCB is used before the distribution board in a wiring circuit?**
 An RCCB (Residual Current Circuit Breaker) is placed before the distribution board to provide **earth leakage protection** for the entire installation. It disconnects the supply immediately if it detects current leakage to the earth, preventing electric shocks and fire hazards.
18. **Name two distinct functions of Relay.**
 Two distinct functions of a relay are:
- **Fault Detection:** Sensing abnormal conditions (like over-current or over-voltage) in the circuit.
 - **Circuit Isolation Control:** sending a trip signal to the Circuit Breaker to isolate the faulty section of the system.

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Industrial Electrical System Module 1 Mathematical Questions

Mathematical Questions

- (a) **How much voltage is applied for insulation testing between live conductor and earth for a Low Voltage installation?**
 For low voltage installations, a test voltage of **500 Volts DC** is applied. The insulation resistance should be not less than 1 Mega-ohm.
- (b) **How much clear space is required in front of a switch board?**
 According to the Indian Electricity Rules (Rule 51), a clear space of not less than **1 meter** (approx 3 feet) in width is required in front of the switchboard.
- (c) **How much horizontal clearance for low/medium voltage lines adjacent to a building?**
 For low and medium voltage lines passing adjacent to a building, the minimum horizontal clearance required is **1.2 meters** (approx 4 feet) from the nearest point.
- (d) **Maximum distance between supporting pillars for medium voltage overhead line?**
 The standard maximum span (distance) for medium voltage overhead lines is typically **65 meters** (approx 220 feet) for urban distribution, though this varies by pole type (e.g., 50-80m for steel poles, 80-100m for RCC).
- (e) **Maximum number of service lines from one pillar of overhead line?**
 The standard practice is to tap a maximum of **4 service lines** from a single overhead line support (pillar) to avoid mechanical overcrowding.

- (f) **If actual wattage is not indicated, what is considered for ceiling fan while calculating load?**

When the actual wattage is not specified, the standard load assumed for a ceiling fan is **60 Watts** (per modern National Electrical Code standards).

5 Marks Questions

Theory Questions

- (a) **Identify the reason why Cut off and Energy Meter are sealed by the supplier of Electric power.**

The utility supplier seals the cut-out (fuse) and the energy meter for two primary reasons:

- **Prevention of Power Theft:** To ensure that consumers cannot bypass the meter or tamper with the internal connections to slow down or stop the recording of energy consumption.
- **Safety and Integrity:** To prevent unauthorized access to live parts on the supply side, ensuring that only authorized personnel can modify the service connection, thereby maintaining the safety and integrity of the grid supply.

- (b) **Illustrate reason for Star labelling on Air Conditioners and Refrigerators.**

Star labeling, mandated by the Bureau of Energy Efficiency (BEE), indicates the **energy efficiency** of an appliance.

- **Purpose:** It allows consumers to compare the energy consumption of different models.
- **Meaning:** More stars (maximum 5) indicate higher efficiency and lower power consumption for the same output (cooling capacity).
- **Benefit:** It helps in reducing electricity bills and the overall demand on the power grid.

- (c) **Who publishes National Electrical Code? Purposes of this Code.**

Publisher: The National Electrical Code (NEC) of India is published by the **Bureau of Indian Standards (BIS)** (SP 30: 2011).

Purposes:

- To regulate the installation of electrical wiring and equipment for safety.
- To prevent hazards like electrical shocks and fires.
- To standardize practices for electrical engineers, contractors, and inspectors across the country.

- (d) **Compare Wiring diagram and Circuit diagram.**

Wiring Diagram	Circuit (Schematic) Diagram
Shows physical layout and actual position of components.	Shows electrical logic and flow of current.
Used for installation and physical troubleshooting.	Used for understanding system operation and design analysis.
Shows cabling routes and bundles.	Uses standard symbols; physical distance is ignored.

- (e) **Compare merits/demerits of Joint box and Loop-in system.**

- **Joint Box System:**

– *Merit:* Saves wire quantity; lower initial material cost.

- *Demerit:* Joints are potential weak points; fault finding is difficult as joints may be inaccessible (under floors/in roofs).

- **Loop-in System:**

- *Merit:* No hidden joints; easy troubleshooting; faults are accessible at switch/light terminals.
- *Demerit:* Requires more wire/cable length; boxes may become crowded with cables.

(f) **Reason for Distribution box, classify types.**

Reason: A Distribution Box (DB) is used to divide the main power supply into subsidiary circuits (lighting, power, AC, etc.) while providing individual protection (MCB/Fuse) and isolation for each circuit.

Types:

- Based on Construction:** Flush mounted (concealed) or Surface mounted.
- Based on Phase:** SPN (Single Phase Neutral) or TPN (Three Phase Neutral).
- Based on Arrangement:** Horizontal or Vertical DBs.

(g) **Explain functions performed by a MCB.**

A Miniature Circuit Breaker (MCB) performs three distinct functions:

- Overload Protection:** Trips via a bi-metallic strip mechanism when current slightly exceeds the rated limit for a sustained period.
- Short Circuit Protection:** Trips instantly via a magnetic solenoid mechanism during high fault currents.
- Switching:** Acts as a manual switch to turn the circuit ON or OFF for maintenance.

(h) **Parameters to consider before selecting a MCB.**

- **Rated Current (I_n):** The maximum current it can carry continuously (e.g., 6A, 10A, 16A).
- **Rated Voltage:** Operational voltage (230V / 415V).
- **Breaking Capacity:** The maximum short circuit current it can safely interrupt (e.g., 10kA).
- **Tripping Curve/Type:** Type B (Resistive), Type C (Inductive/Motors), or Type D (High inrush).
- **Number of Poles:** SP, DP, TP, TPN, or 4P.

(i) **Differences between MCB and MCCB.**

MCB (Miniature Circuit Breaker)	MCCB (Moulded Case Circuit Breaker)
Rated current up to 100A.	Rated current from 100A up to 2500A+.
Tripping current is fixed.	Tripping current is often adjustable.
Low interrupting capacity (up to 10-15kA).	High interrupting capacity (10kA - 100kA).
Used in domestic/small commercial loads.	Used in industrial/large commercial loads.

(j) **Necessity of test button in RCCB.**

The test button connects a specific resistance across the phases to create a deliberate, small leakage current.

- It verifies that the internal tripping mechanism is mechanically free and functioning correctly.
- It is a mandatory maintenance check to ensure the device will actually trip in the event of a real shock hazard.

(k) **Necessity of RCCB in wiring system.**

An RCCB (Residual Current Circuit Breaker) is necessary for:

- **Human Safety:** It detects earth leakage currents (30mA) which MCBs cannot detect, preventing fatal electric shocks.
- **Fire Safety:** It detects lower-level leakage currents (100mA/300mA) that could generate heat and cause electrical fires over time.

(l) **Technology that RCCB uses, explain.**

RCCB works on the principle of **Kirchhoff's Current Law** using a **CBCT (Core Balance Current Transformer)**.

- Under normal conditions, the current flowing in (Phase) equals the current flowing out (Neutral). The vector sum is zero, so no flux is generated in the toroid.
- During an earth fault, some current leaks to earth. Therefore, $I_{phase} \neq I_{neutral}$.
- This difference creates a residual magnetic flux in the core, inducing a voltage in the sensing coil, which activates the tripping relay and opens the contacts.

(m) **Requirement of Relay in control circuit.**

A relay is an electromechanical switch used to:

- **Circuit Isolation:** Isolate the low-voltage control circuit (e.g., 24V DC) from the high-voltage power circuit (e.g., 230V AC).
- **Automation:** Allow a small electronic signal to switch a large load.
- **Logic Control:** Enable complex switching logic (latching, timing, sequencing) in control panels.

Mathematical Questions

(a) **Connected load calculation with diversity (room with tube lights, fan, plug points), recommend cable size.**

Example Calculation:

Assume a room contains:

- 2 Tube lights (40W each)
- 2 Ceiling Fans (60W each)
- 2 Plug points (rated 100W each)

Total Connected Load:

$$P = (2 \times 40) + (2 \times 60) + (2 \times 100) = 80 + 120 + 200 = 400 \text{ Watts}$$

Current Calculation:

$$I = \frac{Power}{Voltage} = \frac{400}{230} \approx 1.74 \text{ Amps}$$

Diversity Factor: For lighting circuits in a single room, diversity is often considered 1.0 (all on), but even if 0.8 is applied, the current remains low ($< 2A$).

Recommended Cable:

- Minimum standard for lighting circuits: **1.5 sq. mm Copper cable**.
- Protection: 6A or 10A MCB (C-Curve).

(b) **Cable size for one power circuit (2 kW Water Heater + 16 A socket outlet).**

Load Calculation:

- Heater Load: 2000 Watts
- Voltage: 230 Volts

Current Calculation:

$$I = \frac{2000}{230} \approx 8.7 \text{ Amps}$$

However, the circuit includes a **16A Socket**, which is rated to handle up to 16 Amps (approx 3600W) depending on what is plugged in. Design should be based on the rating of the socket or the maximum likely load.

Design Current: 16 Amps.

Recommended Cable:

- To safely carry 16A continuously and account for starting currents or voltage drop: **2.5 sq. mm or 4.0 sq. mm Copper cable**.
- **4.0 sq. mm** is preferred for heaters to minimize heating in the wire.
- Protection: 16A or 20A MCB.

10 Marks Questions

Mathematical Questions

- (a) **Calculate gross amount of power bill (complex tariff).**

Note: Since specific consumption data was not provided, the following is a solved example based on a typical Industrial Two-Part Tariff.

Problem Statement:

An industrial consumer has a connected load of 100 kW. The Maximum Demand recorded is 80 kVA. The units consumed (Energy) in a month are 15,000 kWh. Calculate the monthly electricity bill based on the following tariff structure:

- **Demand Charge:** Rs. 300 per kVA of Maximum Demand per month.
- **Energy Charge:** Rs. 6.50 per kWh (Unit).
- **Power Factor Surcharge:** Nil (Assume unity PF or included in kVA).
- **Electricity Duty:** 5% of the total energy and demand charges.

Solution:

Step 1: Calculate Demand Charges (Fixed Charges)

$$\text{Demand Charge} = \text{Max Demand (kVA)} \times \text{Rate per kVA}$$

$$\text{Demand Charge} = 80 \times 300 = \text{Rs. } 24,000$$

Step 2: Calculate Energy Charges (Running Charges)

$$\text{Energy Charge} = \text{Units Consumed (kWh)} \times \text{Rate per kWh}$$

$$\text{Energy Charge} = 15,000 \times 6.50 = \text{Rs. } 97,500$$

Step 3: Calculate Total Base Bill

$$\text{Base Bill} = \text{Demand Charge} + \text{Energy Charge}$$

$$\text{Base Bill} = 24,000 + 97,500 = \text{Rs. } 1,21,500$$

Step 4: Calculate Electricity Duty

$$\text{Duty} = 5\% \text{ of Base Bill}$$

$$\text{Duty} = 0.05 \times 1,21,500 = \text{Rs. } 6,075$$

Step 5: Calculate Gross Amount

$$\text{Gross Amount} = \text{Base Bill} + \text{Duty}$$

$$\text{Gross Amount} = 1,21,500 + 6,075 = \text{Rs. } \mathbf{1,27,575}$$

Theory Questions

- (a) **Justify and compare Switch fuse unit vs MCB for induction motor. Comparison:**

- **Protection Sensitivity:** MCBs (Motor protection type) are more sensitive to small overloads compared to rewirable fuses in a Switch Fuse Unit (SFU).
- **Single Phasing:** A major risk for motors is "single phasing" (loss of one phase). A fuse may blow in only one phase, leaving the motor running on two phases, which can burn the motor windings. A TPN MCB trips all three phases simultaneously if an internal fault mechanism detects an issue, offering better safety.
- **Restoration:** MCBs can be reset immediately after a trip. Fuses require replacement/rewiring, increasing downtime.

Justification: For smaller motors (e.g., up to 10HP), an **MCB (Type-C or Motor Protection Circuit Breaker - MPCB)** is preferred due to ease of maintenance and simultaneous pole tripping. For very large motors, specialized protection relays with fuses (for high rupture capacity) or MCCBs are used, but the MCB is superior to a simple rewirable SFU for general motor safety.

- (b) **MCB vs MCCB recommendation for workshop, and justification.**

Recommendation: MCCB (Moulded Case Circuit Breaker) is recommended for the main incomer of an industrial workshop, while MCBs are suitable for individual small sub-circuits.

Justification:

- **Current Rating:** Workshops often require main currents exceeding 100A (e.g., 125A, 250A), which is beyond the range of standard MCBs (usually max 63A or 100A). MCCBs handle currents up to 2500A.
- **Adjustability:** Workshop loads vary. MCCBs offer adjustable thermal and magnetic trip settings, allowing the engineer to fine-tune protection based on the actual workshop load profile.
- **Breaking Capacity:** Industrial faults can generate high short-circuit currents (e.g., 25kA). MCBs usually handle only up to 10kA. MCCBs can handle much higher fault currents, ensuring the breaker doesn't explode during a major fault.

- (c) **Constructional features and function of MCB, working principle and application of RCCB.**

A. MCB (Miniature Circuit Breaker)

- **Construction:**
 - i. *Bimetallic Strip:* For overload sensing (thermal operation).
 - ii. *Solenoid Coil:* For short-circuit sensing (magnetic operation).
 - iii. *Arc Chute:* A stack of metal plates to split and cool the arc.
 - iv. *Operating Mechanism:* Spring-loaded latch for quick make/break.
- **Function:** Automatically switches off the electrical circuit during abnormal conditions like overload and short circuit.

B. RCCB (Residual Current Circuit Breaker)

- **Working Principle:** Based on Kirchhoff's Current Law. It uses a Core Balance Current Transformer (CBCT). Under normal conditions, the phase current (I_p) equals neutral current (I_n), and the vector sum is zero. During an earth fault, leakage current flows to earth, so $I_p \neq I_n$. This imbalance creates a magnetic flux in the toroidal core, inducing voltage in a sensing coil which activates the relay to trip the switch.
- **Application:** Used in domestic and industrial distribution boards specifically to prevent electric shocks (human safety) and earth-leakage fires.

(d) Main structural parts of an Energy meter.

The induction type energy meter consists of four main systems:

- i. **Driving System:** Consists of two electromagnets. The *Shunt Magnet* (Pressure Coil) is connected across the supply voltage, and the *Series Magnet* (Current Coil) is connected in series with the load.
- ii. **Moving System:** A light aluminum disc mounted on a vertical spindle. It rotates due to the torque produced by the interaction of eddy currents and magnetic fluxes.
- iii. **Braking System:** A permanent magnet (C-shaped) placed near the edge of the disc. It induces eddy currents that oppose the motion (Lenz's Law) to control the speed of the disc proportional to power.
- iv. **Registering System:** A train of gears and a dial or cyclometer mechanism that counts the rotations of the disc and displays the energy consumed in kWh.

(e) What is relay, types of relay, overload protection for induction motor.

Relay: A sensing device that detects a fault (abnormal condition) in the system and sends a trip command to the circuit breaker to isolate the faulty part.

Types of Relays (based on construction/operation):

- Electromagnetic Attraction type
- Electromagnetic Induction type
- Thermal Relay
- Static / Electronic / Microprocessor-based Relay

Overload Protection for Induction Motor:

The most common protection is the **Thermal Overload Relay**. It works on the heating principle using bimetallic strips. If the motor draws excess current for a prolonged period, the strips heat up and bend, activating the trip mechanism. Modern systems may use microprocessor-based Motor Protection Relays (MPR) which model the thermal capacity of the motor electronically.

(f) Compare merits/demerits of MCB vs re-wirable fuse.

Parameter	MCB (Miniature Circuit Breaker)	Re-wirable Fuse (Kit-Kat)
Sensitivity	High. Trips quickly on slight overloads.	Low. Relies on wire melting, which is slow.
Restoration	Easy. Just toggle the switch ON.	Difficult. Requires replacing the fuse wire.
Safety	High. No live parts exposed.	Low. Risk of shock while replacing wire.
Cost	Higher initial cost.	Very low initial cost.
Calibration	Factory set and accurate.	Inaccurate. Depends on the wire used by the electrician.
Aging	Performance is stable over time.	Wire oxidizes over time, affecting rating.